

Fariha Ansari

Lecturer - Mathematics | PhD Researcher in Mathematics
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PROFESSIONAL PROFILE

Experienced Mathematics Lecturer with nearly 18 years of continuous university-level teaching and academic service at Sir Syed University of Engineering & Technology (SSUET). My academic profile combines extensive classroom teaching, student counselling, Outcome-Based Education (OBE), assessment design, academic quality assurance, digital learning coordination, and departmental service. I have taught mathematics to more than 1,000 undergraduate students across engineering and multidisciplinary programs and have contributed to developing analytical reasoning, mathematical problem-solving, and concept-based learning among my students. I am currently pursuing a PhD in Mathematics, having completed my doctoral coursework with a CGPA of 3.70/4.00. My approved doctoral research integrates graph theory, combinatorial optimization, scheduling, and machine learning.

EDUCATION

Ph.D. in Mathematics - In Progress | Coursework CGPA: 3.70/4.00

Sir Syed University of Engineering & Technology (SSUET), Karachi | 2022 - Present

Research Topic (Tentative): Optimizing Scheduling Problems using Vertex Coloring and Machine Learning.

- Successfully completed doctoral coursework and advanced to the research phase.
- Research proposal/topic successfully presented and approved; research actively in progress; manuscript in preparation.

Research Focus: graph theory, vertex coloring, combinatorial optimization, scheduling problems, and machine-learning-assisted mathematical optimization.

Supervisor: Dr. Muhammad Naseem, Associate Professor & Chairperson, Department of Software Engineering, SSUET.

M.S. in Mathematics - CGPA: 3.84/4.00

Sir Syed University of Engineering & Technology (SSUET), Karachi | 2013

- Postgraduate study in Mathematics with strong emphasis on advanced mathematical concepts, analytical reasoning, and theoretical foundations.
- Completed the degree with a CGPA of 3.84/4.00.

M.Sc. in Mathematics - First Division

University of Karachi, Karachi | 2007

- Completed postgraduate studies in Mathematics with First Division.

B.Sc. (Hons.) - Mathematics, Chemistry & Physics - First Division

University of Karachi, Karachi | 2006

- Completed honours-level undergraduate studies with Mathematics as the principal academic discipline and supporting study in Chemistry and Physics.

EMPLOYMENT HISTORY

Lecturer of Mathematics

Department of Mathematics and Sciences, Sir Syed University of Engineering & Technology (SSUET), Karachi | 2012 - Present

- Deliver undergraduate mathematics courses to students from engineering, computing, technology, and multidisciplinary programs.
- Plan and deliver concept-driven lectures with emphasis on mathematical reasoning, structured problem solving, and application of mathematics to engineering contexts.
- Develop course material, assessments, and learning activities aligned with institutional academic requirements and Outcome-Based Education (OBE).
- Contribute to course learning outcomes, assessment strategies, academic documentation, and continuous improvement activities.
- Provide regular academic guidance, student counselling, and support for students with different levels of mathematical preparation.
- Use digital learning environments and technology-enabled teaching practices to support course delivery and student engagement.

Junior Lecturer of Mathematics

Department of Mathematics and Sciences, Sir Syed University of Engineering & Technology (SSUET), Karachi | 2008 - 2012

- Supported the delivery of foundational and undergraduate mathematics courses.
- Assisted with assessment, academic documentation, student support, and departmental academic operations.
- Developed early teaching experience in communicating mathematical concepts to engineering students and supporting their transition to university-level mathematics.

ACADEMIC & ADMINISTRATIVE RESPONSIBILITIES

Departmental Coordinator - Virtual Learning Environment (VLE)

Sir Syed University of Engineering & Technology (SSUET) | 2019 - Present

- Coordinate departmental use of the Virtual Learning Environment and support technology-enabled academic delivery.
- Facilitate faculty and student engagement with VLE/LMS systems and digital teaching practices.
- Support continuity and organization of digital course delivery, particularly in technology-mediated learning environments.

Member - Self-Assessment Report (SAR) Committee

Sir Syed University of Engineering & Technology (SSUET) | 2015 - 2024

- Contributed to academic quality assurance and self-assessment activities within the department.
- Supported preparation and maintenance of accreditation-related academic documentation, evaluation evidence, and continuous-improvement records.
- Participated in departmental processes related to course assessment, academic standards, and quality enhancement.

Ramadan Drive - Organizer & Volunteer

Sir Syed University of Engineering & Technology (SSUET) | 2024 & 2025

- Organized and actively participated in SSUET Ramadan Drive initiatives.
- Contributed to coordination, organization, and community-service activities associated with the drives.

Student Academic Support & Counselling

Sir Syed University of Engineering & Technology (SSUET) | Ongoing

- Provide academic counselling and guidance to undergraduate students alongside formal teaching responsibilities.
- Support students in strengthening mathematical foundations, analytical thinking, and problem-solving approaches.

TEACHING PROFILE

Extensive experience teaching mathematics across engineering and multidisciplinary undergraduate cohorts. Teaching exposure includes:

- Calculus & Analytical Geometry
- Differential Equations
- Complex Variables & Integral Transforms
- Discrete Mathematics / Discrete Structures
- Multivariable Calculus
- Linear Algebra & Differential Equations
- Linear Algebra & Vector Analysis
- Numerical Methods for Engineers
- Delivered lectures to 1000+ undergraduate students across engineering disciplines.
- Promotes critical thinking, analytical reasoning, mathematical communication, and systematic problem solving.
- Integrates theoretical foundations with engineering-oriented examples and practical mathematical applications.
- Experienced in course delivery, assessment preparation, student evaluation, and academic counselling.

RESEARCH PROFILE & INTERESTS

Current Doctoral Research (Tentative): Optimizing Scheduling Problems using Vertex Coloring and Machine Learning.

Research Status: Doctoral coursework completed; research topic successfully proposed and approved; research actively in progress; manuscript in preparation.

Research Orientation: The work brings together mathematical optimization and computational approaches, with particular emphasis on graph-theoretic modeling of scheduling problems and the potential use of machine learning to improve optimization and decision-making processes.

- Graph Theory and Vertex Coloring
- Combinatorial Optimization
- Scheduling and Resource-Allocation Problems
- Machine Learning Applications in Mathematical Optimization
- Computational Mathematics
- Mathematical Modeling and Algorithmic Problem Solving

OUTCOME-BASED EDUCATION & QUALITY ASSURANCE

- Experience contributing to OBE-aligned course delivery, assessment strategies, course learning outcomes, and academic evaluation processes.
- Participated in capacity-building activities related to OBE, QOBE, rubrics, complex engineering problems/activities, and implementation of OBE in non-engineering courses.
- Contributed through SAR committee responsibilities (2015-2024) to self-assessment, quality documentation, accreditation support, and continuous academic improvement.
- Experienced in maintaining academic documentation and supporting evidence-based review of teaching and assessment practices.

DIGITAL LEARNING & ACADEMIC TECHNOLOGY

- Departmental VLE Coordinator with sustained responsibility for supporting digital academic delivery since 2019.
- Experience with MS Teams, LMS/VLE platforms, MS Forms, and technology-enabled teaching workflows.
- Participated in institutional training on online class delivery, MS Teams, MS Forms, SSUET Virtual Learning Environment, and MOOCs.
- Supports faculty and students in effective use of digital learning environments for course communication and academic engagement.

FACULTY DEVELOPMENT & KNOWLEDGE-SHARING ACTIVITIES

- **Advanced Pedagogy in Mathematics** - Mathematics Department, SSUET - 28 September 2022.
- **Developing 21st Century Technological Communication Skills** - English Department, SSUET - 22 September 2022.
- **Optimize Your Performance** - Mathematics Department, SSUET - 16 September 2022.
- **Understanding and Importance of MOOC in Learning** - Mathematics Department, SSUET - 15 September 2022.
- **Presented "Critical Thinking"** under the Capacity Building Program, Department of Applied Sciences - 21 January 2021.
- **Do's and Don'ts of Outcome Based Assessment (OBE)** - Capacity Building Program - 25 January 2021.
- **Roles of University in Implementing Sustainable Development** - Capacity Building Program - 25 January 2021.
- **QOBE Software and Rubrics in QOBE** - Capacity Building Program - 15 January 2021.
- **Complex Engineering Problem / Complex Engineering Activity** - Capacity Building Program - 2020-2021.
- **Implementation of OBE-Based System in Non-Engineering Courses** - Capacity Building Program - 2020-2021.
- **MS Teams, MS Forms and SSUET Virtual Learning Environment** - Capacity Building Program - January 2021.
- **Heat Waves and Urban Heat Island** - Sardar Yasin Malik Professional Development Centre, University of Karachi - 17-18 April 2019.
- **Why Not Excel: A Simple and Easy Approach for Statistical Analysis** - ORIC, University of Karachi - 11 September 2018.

ACADEMIC & TECHNICAL COMPETENCIES

- Mathematical Modeling & Analytical Problem Solving
- Graph Theory, Vertex Coloring & Optimization
- Research Methodology & Academic Writing
- Outcome-Based Education (OBE) & Assessment Design
- Academic Quality Assurance & SAR Documentation
- Digital Teaching Tools: MS Teams, LMS, VLE Platforms & MS Forms
- Mathematical & Research Software: MATLAB, Python (NumPy, SciPy), LaTeX & MS Excel
- Critical Thinking, Communication, Student Counselling & Academic Coordination

LANGUAGES

- English - Professional working proficiency
- Urdu - Native / full professional proficiency

REFERENCES

Dr. Muhammad Naseem

Associate Professor & Chairperson, Department of Software Engineering, SSUET, Karachi
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Dr. Asma Zaffar

Associate Professor & Chairperson, Department of Mathematics and Sciences, SSUET, Karachi
Email: aszafar@ssuet.edu.pk

Dr. Nazish Kanwal

Assistant Professor of Mathematics, HEC Approved PhD Supervisor, FAST-NUCES, Karachi Campus
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